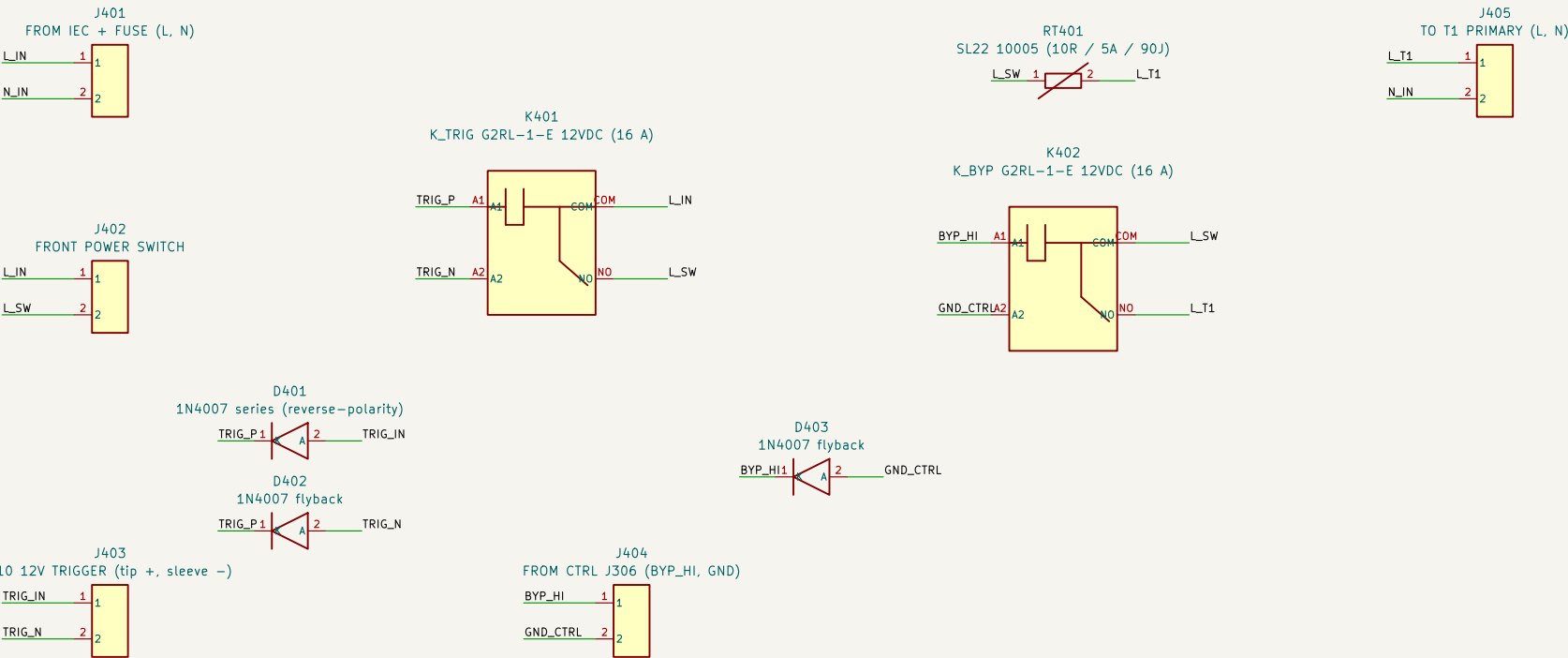


MAINS BOARD V0.6 / FIRST CIRCUIT DRAFT 2026-10-01 / LIVE 110 VAC: creepage >= 6.4 mm between mains and the 12 V coil side, PE to chassis stud (not on this board)

Path (docs/14 2): IEC inlet + fuse holder (chassis) -> J401 -> [manual switch J402 || K_TRIG contact] -> NTC RT401 (K_BYF contact across it) -> J405 -> T1 primary.



K_TRIG coil 33 mA at 12 V from the Z10 trigger output (<= 50 mA budget, Z10 spec unknown). Contact in parallel with the front switch (OR). If the Z10 cannot source 33 mA: latch via the 12 V aux (docs/14 6.7).

NTC: primary 1.79 A rms steady (197 VA label), 10 ohm cold limits the first half-cycle to ~15 A; 90 J rating vs 38 J estimated absorption (docs/14 6.2). Bypassed after ~1.1 s by K402.

If K402 never pulls in, RT401 carries 1.79 A continuously: SL22 10005 is rated 5 A steady, so it survives (hot, ~1 ohm) – the failure is only lost inrush limiting on a quick off-on.

Mains-side parts: J401, J402, J405, RT401, K401/K402 contacts. Low-voltage: K401/K402 coils, D401-D403, J403, J404. Keep >= 6.4 mm creepage; slot under the relays.

V0.6 mains board / circuit draft only / LIVE MAINS / no hardware measurements

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Title: LM3886 V0.6 mains soft-start / trigger board (MAINS) – LIVE MAINS

Size: A3 Date: 2026-10-01 Rev: 0.6 FIRST DRAFT
KiCad E.D.A. 10.0.6 Id: 1/1